Private Equity Diligence Agent

Benchmark Framework, Scoring Methodology & Multi-Agent Architecture

Engineering & Product Specification · Version 2.0

Lower Middle Market ($10–100M EV) · Industry-Agnostic Core + Industry Overlays · Confirmatory Pre-Close Diligence

Primary criteria sourced from Rallyday Partners (Austin Hough). Public-source benchmarks are supplementary.

Executive Summary

This document is the v2 engineering and product specification for the Rallyday Partners PE diligence agent. It incorporates three material changes from v1: (1) primary evaluation thresholds are now sourced directly from Rallyday Partners' investment team — specifically Austin Hough's written guidance on tech services and healthcare services businesses — with public-source benchmarks retained as optional secondary reference; (2) the agent architecture has been restructured into a five-phase, ten-agent pipeline per the agreed system design; and (3) document-priority guidance from the team's working session has been formalized into the Ingestion Agent specification.

The core design philosophy does not change: the agent mirrors how an experienced PE investment professional forms an initial underwriting view, surfaces flags and questions, and hands judgment back to the deal team. It does not produce composite scores, and it does not replace the QofE provider, outside counsel, or sector advisor.

What changed in v2

- **Primary thresholds (Austin).** Austin's email criteria for tech services and healthcare services are now the primary flag thresholds for those two industry overlays. Public benchmarks are noted in each section as secondary reference only.
- **Multi-agent architecture.** The single-agent pipeline from v1 is replaced by a five-phase, ten-agent design. Each agent has its own input contract, output schema, and hand-off specification. Section 0 covers the architecture; Sections 1–9 are restructured accordingly, each mapping to one workstream agent.
- **Document priority tier.** The Ingestion Agent now operates a two-tier document model: Priority Tier (the ~5–10 documents that contain ~90% of relevant signal) is processed first and results surfaced immediately; the remaining documents are ingested in background and available for deep-dive queries. This reflects Austin's explicit guidance on where to start.
- **Pragmatic Phase 1 posture.** For the first implementation phase, the agent treats stated metrics as source of truth. Complex reconciliation (e.g., recomputing NRR from raw cohort data vs. the stated figure) is Phase 2+ work, not Phase 1. This is a deliberate scoping decision, not an oversight.
- **Thresholds are flags, not verdicts.** Austin was explicit: these are directional screens, not hard rules. The system should surface flags for the investment professional to opine on — the investment professional always makes the call. The agent must never block a deal based on a threshold breach.

0. Multi-Agent Architecture

The system is organized into five sequential phases. Within phases, agents can run in parallel where their inputs are independent. The architecture diagram below represents the agreed design from the Rallyday / Nimble Gravity working session.

0.1 Five-Phase Pipeline Overview

Phase	Agent(s)	Runs in parallel?	Primary output
1 · Ingestion	Ingestion Agent	No — must complete before Phase 2	Indexed document store; Priority Tier flagged
2 · Dual Classification	Document Classifier + Company Profiler	Yes — run in parallel	Each doc tagged by workstream; industry overlay confirmed
3 · Parallel Workstream Analysis	Business Model Agent · Customer Quality Agent · Financial Trends Agent · KPI Agent · Legal & Contracts Agent · Quality of Earnings Agent · Forecast Agent	Yes — all seven run in parallel after Phase 2	Structured fact store + per-agent rating + flag list per workstream
4 · Cross-Analysis	Cross-Analysis Agent	No — waits for all Phase 3 outputs	Reconciliation log; ranked red flags; CIM vs. data room mismatch list; top 10 issues
5 · Synthesis & Delivery	Orchestrator Agent	No — waits for Phase 4 output	Final memo, one-pager, risk grid, KPI dashboard, question lists, source citations

0.2 Cross-Cutting Layer

Two capabilities run across all phases and all agents:

- **Citation and provenance.** Every extracted fact carries: source document name, page/tab/cell reference, raw quoted text, and extraction confidence (high / medium / low). This is non-negotiable — it is the primary mechanism that allows the deal team to verify any claim without re-reading the VDR.
- **Human-in-the-loop + Databricks memory.** The deal team can override any rating, add context, or flag a finding as 'known / accepted.' Overrides are stored in the Databricks memory layer and persist across sessions for the same deal. The memory layer also accumulates cross-deal portfolio benchmarks over time to supplement public-source data.

0.3 Document Priority Tier

This is the most operationally important addition in v2. Austin's guidance was unambiguous: five to ten documents contain approximately 90% of the diligence-relevant signal in a typical banked deal. The

Ingestion Agent processes Priority Tier documents first and surfaces results immediately, while the remaining documents are indexed in background.

Priority Tier document hierarchy (in order):

Priority	Document type	Why first	Banked deal?	Non-banked fallback
1	Confidential Information Memorandum (CIM)	If a good banker prepared it, it summarises 90% of the opportunity: financials, customers, KPIs, growth thesis, risks. The Bible of the banked process.	Yes — always exists	Move to Priority 2 or 3
2	QofE Report (if present)	Third-party vetted historical EBITDA. Tells you what the real earnings base is before any addback debate. Historic-facing only — no projections.	Often exists in well-run processes	Skip; surface absence as flag
3	Financial Forecast / Projection Model	Management's projection model. Contains revenue build, opex plan, EBITDA bridge, and often pipeline or backlog data supporting the projection.	Often an Excel model	P&L PDFs if no model
4	KPI / Dashboard File	Internal KPI tracking. If labelled 'KPI' in the filename, it almost always contains the metrics that management uses to run the business — utilization, NRR, pipeline, etc.	Sometimes embedded in CIM	Request explicitly if absent
5	Customer Revenue Workbook	Revenue by customer, historically. Sometimes embedded in the QofE or model, sometimes standalone. Needed for concentration and retention analysis.	Often standalone Excel	Extract from CIM if embedded
6	Pipeline / Backlog File	Forward revenue visibility. Weighted pipeline, conversion assumptions, and backlog. On tech services deals: key to understanding forecast credibility.	Often standalone	Request explicitly
7	Cap Table / Capitalization Table	Ownership structure. Who controls the decision, any unusual equity structures, options/warrants that could complicate close.	Usually an Excel file	Request from founders
8–10	Material customer	Once customer concentration is	Varies	Flag as missing if absent

Priority	Document type	Why first	Banked deal?	Non-banked fallback
	contracts (top by revenue)	identified from the CIM or customer workbook: if any single customer >20–25% of revenue, find and review their contract immediately — CoC consent and termination rights are deal-relevant.		

ℹ️ *For non-banked deals (no CIM, no banker): begin with the financial model and any folder labelled 'Financials.' P&L PDFs, balance sheet PDFs, and tax returns are the starting point. The absence of a CIM is itself a flag — note reduced confidence across all sections.*

0.4 Industry Overlay Model

The Company Profiler (Phase 2) determines the industry overlay. Two overlays have primary thresholds sourced from Rallyday Partners:

- **Tech Services (primary thresholds: Austin Hough email, May 2026).** Covers tech-enabled services, IT services, digital services, consulting with a technology delivery component.
- **Healthcare Services (primary thresholds: Austin Hough email, May 2026).** Covers physician practices, home care, hospice, behavioral health, dental, dermatology, and adjacent regulated care services.

Three additional overlays retain public-source benchmarks as primary (to be refined as Rallyday deal data accumulates):

- B2B SaaS / tech-enabled software
- Industrial / manufacturing / specialty distribution
- Consumer / DTC / CPG

0.5 Threshold Philosophy

Austin's explicit instruction: thresholds are directional flags for the investment professional to opine on — not hard rules that block a deal. Context always matters (end market, revenue model, deal stage, etc.). The agent must:

- Surface every threshold breach as a flag, with the relevant context extracted from the documents.
- Present the flag neutrally — 'X metric is below the Y% threshold, which suggests Z; context: [extracted text].'
- Never render a verdict. The investment professional decides materiality.
- For red flags that require heavy analytical lift, prefer surfacing the question to management over attempting to compute the answer from raw data in Phase 1.

1. Ingestion Agent (Phase 1)

> ℹ️ *Agent type: wraps deterministic tools. Minimal LLM calls — decides only on fallbacks (OCR, format conversion, skip). This is intentionally low-cost.*

1.1 Purpose

The Ingestion Agent is responsible for: (a) connecting to the data room storage (SharePoint, cloud folder, or local directory); (b) traversing the folder structure and inventorying all documents; (c) applying the Priority Tier classification to determine which documents to process immediately vs. in background; (d) applying OCR or format conversion where needed; and (e) writing all documents to the indexed document store for downstream agents.

The Ingestion Agent does NOT perform content analysis. Its scope ends at 'the document is indexed and accessible.' This keeps it fast, cheap, and easy to debug.

1.2 Priority Tier detection rules

The agent applies keyword matching on filename, folder path, and (for PDFs) first-page text to assign Priority Tier status:

Document type	Detection signals (filename / folder / first-page)
CIM	Keywords: 'CIM', 'Confidential Information Memorandum', 'Offering Memorandum', 'Investment Overview', 'OM'. Typically a PDF of 40–200 pages.
QofE Report	Keywords: 'Quality of Earnings', 'QofE', 'Q of E', 'Financial Due Diligence', 'Sell-Side QofE'. Usually PDF from an accounting firm.
Financial Model / Forecast	Extension: .xlsx or .xlsm. Keywords in filename: 'model', 'forecast', 'projection', 'financial model', 'LBO'. Located in folder labelled 'Financial' or 'Financials'.
KPI / Dashboard	Keywords in filename: 'KPI', 'dashboard', 'metrics', 'scorecard', 'operating'. Any extension.
Customer Revenue Workbook	Keywords: 'customer', 'revenue by customer', 'client revenue', 'ARR by customer'. Usually .xlsx.
Pipeline / Backlog	Keywords: 'pipeline', 'backlog', 'weighted pipeline', 'forecast pipeline', 'sales pipeline'. Usually .xlsx or .pdf.
Cap Table	Keywords: 'cap table', 'capitalization', 'equity', 'cap_table'. Usually .xlsx.
Material customer contract	After customer concentration is identified downstream, the Legal & Contracts Agent will request specific contracts. The Ingestion Agent flags any PDF in a folder labelled 'Contracts', 'Customer Agreements', or 'MSAs'.

1.3 Fallback decisions (the only LLM calls in Phase 1)

- OCR: if pdftotext returns empty or <10 words on a PDF page, route to OCR pipeline.

- Format conversion: .doc → .docx, .xls → .xlsx via LibreOffice wrapper before indexing.
- Skip: files below 1KB, image-only files without OCR-extractable text, duplicate filenames in the same folder.
- Date resolution: if multiple versions of the same document exist (same filename, different modification date), flag the conflict and process the most recent.

1.4 Output contract

Output field	Type	Description
document_id	String (UUID)	Stable identifier for the document across all agents
filename	String	Original filename
folder_path	String	Full folder path within the data room
priority_tier	Boolean	True if Priority Tier; False if background
priority_reason	String	Which detection signal triggered Priority Tier
format	Enum: pdf, xlsx, docx, pptx, other	Post-conversion format
page_count / sheet_count	Integer	For downstream chunking decisions
mod_date	ISO date	Last modified date from file metadata
ocr_applied	Boolean	Whether OCR was used
extraction_confidence	Enum: high, medium, low	high = native text; medium = OCR clean; low = OCR noisy
indexed_text_path	String	Path to extracted text in document store

2. Document Classifier & Company Profiler (Phase 2)

ℹ️ Both agents run in parallel after Phase 1 completes. The Document Classifier assigns workstream tags; the Company Profiler infers the industry overlay. Neither agent generates final analysis — they route documents and configure downstream agents.

2.1 Document Classifier

Takes the indexed document store from Phase 1 and assigns each document to one or more workstream categories. A document can belong to multiple categories (e.g., a QofE report is tagged Financial AND Quality of Earnings).

Workstream tag	Assigned to	Key signals
FINANCIAL	Financial Trends Agent (Guideline 3)	P&L, balance sheet, cash flow, tax returns, audited financials, monthly management accounts
CUSTOMER	Customer Quality Agent (Guideline 2)	Customer lists, revenue by customer, cohort analyses, churn schedules, NRR schedules
KPI_OPS	KPI Agent (Guideline 4)	KPI dashboards, utilization reports, headcount files, pipeline/backlog, operational metrics
LEGAL	Legal & Contracts Agent (Guideline 5)	Contracts, MSAs, SOWs, leases, employment agreements, litigation files, IP documents
QUALITY_EARNINGS	Quality of Earnings Agent (Guideline 6)	QofE reports, EBITDA bridges, addback schedules, revenue reconciliations
FORECAST	Forecast Agent (Guideline 7)	Financial models, projection files, management presentations with forward guidance
BUSINESS_MODEL	Business Model Agent (Guideline 1)	CIM, investor presentations, management decks, org charts, product/service descriptions

Documents that do not match any workstream tag (e.g., individual employee contracts, payroll records, insurance certificates) are classified as BACKGROUND and held in the document store for on-demand query. They are not routed to any workstream agent in Phase 3.

2.2 Company Profiler

Reads the CIM (Priority Tier) and any KPI file to infer the industry overlay. Output is a company profile that configures all downstream agents.

Profile field	Source	How used downstream
Industry overlay	CIM description, service/product terminology, customer verticals	Selects threshold set for all scoring agents
Revenue model	CIM revenue section,	Business Model Agent primary output; drives KPI selection

Profile field	Source	How used downstream
	contract summaries	
Company size indicators	Revenue, headcount, EBITDA from CIM or financial summary	Calibrates benchmark band (e.g., $5–20M ARR SaaS vs. $20–50M)
Deal type	CIM or Nimble Gravity input at kickoff: buyout vs. growth equity	Adjusts emphasis in Forecast Agent and Cross-Analysis Agent
Banked vs. non-banked	Presence or absence of CIM	Adjusts document priority routing and confidence level across all agents
Vertical / sub-sector	CIM first page, customer list sector tags	Selects sub-overlay within healthcare or tech services

3. Business Model Agent (Phase 3 · Guideline 1)

Primary source: CIM. Secondary: org chart, sales playbook, customer contracts. This agent configures industry overlay selection and drives KPI choices in downstream agents.

3.1 Inputs

- All documents tagged BUSINESS_MODEL by the Document Classifier.
- Company profile from the Company Profiler (overlay already set).

3.2 What to extract

Data point	Extraction approach
Revenue model classification	Tag as: pure recurring (subscription/SaaS), repeat-services / retainer, project-based, transactional, usage-based, licensing, marketplace, hybrid (with %). Extract the percentage split where stated.
Products / services by line	Revenue split, gross margin per line (if stated), and growth rate per line.
Customer segments / verticals	Tag by end-market vertical and customer size segment (enterprise / mid / SMB).
Sales motion	Tag as: founder-led, enterprise sales, channel/partner, inbound/PLG, outbound, marketplace. Note if transitioning.
Revenue visibility	Contracted revenue as % of forward 12-mo plan. Backlog coverage ratio (months).
Key dependencies	Any vendor/platform/channel >10% of revenue or COGS.
Recent model changes	Pricing changes, GTM changes, customer-mix changes in trailing 24 months. Flag each.

3.3 Revenue model quality flag

The one rating this agent produces — a proxy for revenue durability:

Flag	Trigger
Green	≥70% recurring or contracted, OR repeat-rate >80% with multi-year tenure
Yellow	40–70% recurring/contracted, or repeat-services with informal but strong customer relationships
Red	<40% recurring/contracted AND no demonstrated repeat-rate; OR revenue is mostly one-time project work (Austin: flag explicitly for tech services)

Austin (tech services): 'Revenue that is mostly one-time project work: not necessarily a deal-killer, but it usually deserves a flag.' This aligns with the Red trigger above.

3.4 Output contract

- Revenue model tag + durability flag (Green/Yellow/Red) + confidence.
- Structured company description paragraph (2–3 sentences).
- Industry overlay confirmation (or conflict flag if Profiler and Business Model Agent disagree).
- List of data room gaps relevant to this section (e.g., no org chart, no sales playbook).

4. Customer Quality Agent (Phase 3 · Guideline 2)

Primary sources: customer concentration schedule, customer revenue workbook, cohort analysis (if present). Secondary: NRR schedules, churn logs, contract samples.

4.1 Primary thresholds — Tech Services (Austin, primary)

Metric	Flag threshold	Austin's note
Net Revenue Retention	**Red: <~90%**	Suggests churn pressure, weaker customer stickiness, or a more project-based / transactional model
Gross Revenue Retention	**Red: <~80–85%**	Austin's explicit threshold
Top customer concentration	**Red: >~25% of revenue**	Should be flagged as a concentration risk, particularly if project-driven, contractually uncommitted, etc.
Average account size	**Yellow: <~$100–200K**	Smaller accounts may imply more SMB exposure, lower budget durability, and a harder path to scale

Secondary reference (public): single customer >15% is a common PE flag; SBA lenders often knock out at >20%. Austin's threshold of ~25% is the primary standard for this client.

4.2 Primary thresholds — Healthcare Services (Austin, primary)

Metric	Flag threshold	Austin's note
Customer / referral source concentration	**Red: any referral source, facility, health system, payor, or customer >~20–25% of volume or revenue**	Treat like customer concentration in tech — flag at same level
Government / payor concentration	**Yellow: >~50% Medicare, Medicaid, VA, or small number of managed care / government-related payors**	Worth flagging if heavily dependent on government reimbursement

4.3 What to extract

Data point	Source	Phase 1 posture
Top 10 customers by revenue (3-yr history)	Customer concentration schedule, CIM	Extract and present as stated; do not recompute
NRR (if stated)	CIM, QofE, KPI file	Extract stated figure; flag if methodology not explained
GRR (if stated)	Same	Extract stated figure
Logo churn rate (annual)	Customer list YoY, CIM	Extract if stated; flag if absent

Data point	Source	Phase 1 posture
Customer tenure distribution	Customer master file, CIM	Extract average tenure if stated
Average account size	Revenue ÷ customer count if both stated	Compute only if both inputs are directly stated
Payor mix breakdown (healthcare)	QofE, management reports, CIM	Extract percentage split as stated

4.4 Contract risk — CoC and termination (trigger for Legal Agent)

Austin specifically called this out during the working session: once a material customer (>20–25% of revenue) is identified, immediately check whether their contract is in the VDR and whether it contains change-of-control consent or termination-for-convenience provisions. This is the bridge between the Customer Quality Agent and the Legal & Contracts Agent.

- **Trigger:** Any customer identified as >20% of revenue generates an automatic request to the Legal & Contracts Agent to locate and review that customer's contract.
- **Flag if missing:** If no contract is found for a customer >20% of revenue, this is a data room gap — flag as high priority information request.

4.5 Output contract

- Concentration table: top 10 customers by revenue %, trend, flag per customer.
- Retention summary: stated NRR / GRR / logo churn + flag vs. primary threshold.
- Contract trigger list: customers >20% with contract review request to Legal Agent.
- Data room gaps: any expected customer data not present.

5. Financial Trends Agent (Phase 3 · Guideline 3)

Primary sources: audited financials, monthly P&L management accounts, EBITDA bridge. Secondary: tax returns, board materials. Phase 1 posture: extract stated figures as source of truth. Cross-document reconciliation is Phase 2+ work handled by the Cross-Analysis Agent.

5.1 Primary thresholds — Tech Services (Austin, primary)

Metric	Flag threshold	Austin's note
Organic revenue growth	**Red: <~10–15%**	For a premium tech services platform, low growth raises questions around market positioning, sales engine maturity, customer demand
Gross margin	**Red: <~40%**	Could indicate lower-value work, weak utilization, poor delivery leverage, or insufficient pricing power. Premium digital services: 45–50%+ preferred
EBITDA margin	**Yellow: <~10–15%**	Depends on company stage — may be fine, but should be flagged

Secondary reference (public): services/consulting gross margin typically 35–55%; EBITDA 15–25% for mature platforms. Austin's 40% gross margin threshold is primary for this client.

5.2 Primary thresholds — Healthcare Services (Austin, primary)

Metric	Flag threshold	Austin's note
Revenue growth (same-store)	**Red: <~5–10%**	Low growth may be fine in certain stable markets, but should prompt questions around referral trends, reimbursement pressure, competition, or provider / caregiver availability
Gross margin	**Red: <~30–35%**	For people-heavy healthcare services: key indicator of wage pressure, poor labor utilization, pricing constraints, or unfavorable payor / customer mix
EBITDA margin	**Red: <~10%**	Varies by subsector — sub-10% can indicate labor inefficiency, weak reimbursement, poor scheduling / utilization, high admin burden, or lack of scale
Revenue tied to episodic / event-driven demand	**Yellow: flag if volume is hard to forecast, referral patterns are inconsistent, or revenue depends on discrete events**	Similar to project revenue in tech — flag if not recurring

5.3 What to extract

Category	Specific metrics	Phase 1 posture
Revenue trend	Annual revenue for each year available; YoY growth rate; TTM if stated	Extract as stated
Gross margin	Stated gross margin %; gross profit dollars by period	Extract as stated; compute % if both revenue and GP stated
EBITDA / EBITDA margin	Reported EBITDA; mgmt-adjusted EBITDA; EBITDA margin	Extract all versions stated; do not recompute in Phase 1
Revenue by line / segment	If stated: revenue split by product, service line, geography, or customer segment	Extract as stated
Cost structure highlights	If stated: key cost categories, headcount cost as % of revenue, fixed vs. variable split	Extract narrative + any stated %
Working capital indicators	DSO, DPO if stated; AR aging if present	Extract stated figures
Budget vs. actual	If board materials present: stated vs. actual variance for revenue and EBITDA	Extract and flag any material misses

5.4 EBITDA addback awareness

The Financial Trends Agent extracts the addback schedule if present and passes it to the Quality of Earnings Agent. It does not evaluate addbacks — that is QofE Agent scope. It does flag:

- Total addbacks as % of reported EBITDA — if >20%, flag for QofE Agent priority.
- Whether an addback bridge is present at all — if absent, flag as data room gap.

5.5 Output contract

- Revenue trend table (year by year, growth rate, flag vs. threshold).
- Margin summary (gross and EBITDA, flag vs. threshold, trend direction).
- Working capital snapshot (DSO/DPO if available).
- Addback passthrough to Quality of Earnings Agent.
- Data room gaps (e.g., no monthly P&L, no audited financials for last year).

6. KPI Agent (Phase 3 · Guideline 4 · Uses Vertical)

ℹ️ This is the most overlay-dependent agent. The KPI set is entirely different between tech services, healthcare, SaaS, industrial, and consumer. The industry overlay from the Company Profiler is a required input.

6.1 Primary thresholds — Tech Services (Austin, primary)

KPI	Flag threshold	Austin's note
Contractor-heavy or single-geography delivery model	**Yellow: flag if contractor-heavy or if delivery concentrated in India or other less-desirable geographies**	Especially in India or less desirable geographies — delivery model risk
Average account size	**Yellow: <~$100–200K ACV**	Smaller accounts imply SMB exposure, lower durability, harder to scale
Utilization (where trackable)	Green: ≥75%; Yellow: 65–75%; Red: <65%	Secondary / public benchmark — Austin did not specify directly; use if data available
Backlog / pipeline coverage	**Yellow: <6 months of revenue in backlog or qualified pipeline**	Key to forecast credibility on tech services deals

6.2 Primary thresholds — Healthcare Services (Austin, primary)

KPI	Flag threshold	Austin's note
Employee / caregiver turnover	**Red: >~30–40%**	High turnover creates quality issues, hiring costs, capacity constraints, compliance risk, and margin pressure
Clinician / caregiver recruiting funnel	**Yellow: flag if constrained recruiting caps growth**	Even if current revenue looks good, constrained recruiting can cap growth
Utilization / productivity	**Yellow: flag low utilization for clinic-based, provider-based, home care, or field services models**	Major margin driver — flag if utilization data absent (it should be tracked)
Compliance / quality issues	**Red: any history of audits, adverse survey findings, licensing issues, litigation, credentialing gaps, or billing / coding concerns**	Any history — not just open issues
Site / branch-level visibility	**Yellow: flag if multi-site company cannot produce revenue, margin, volume, utilization, and quality metrics at the location level**	If they cannot produce this, that is an issue — management capability flag

6.3 KPI extraction by overlay

The KPI Agent extracts the applicable KPI set based on the confirmed overlay. If a KPI the agent expects to see is not trackable from the documents, that absence is itself a flag.

Overlay	KPIs to extract
Tech Services	Utilization rate, bill rates by role, contractor vs. employee split, average account size / ACV, bookings, backlog, pipeline coverage, revenue per FTE, attrition
Healthcare Services	Census / patient panel, payor mix %, caregiver / clinician headcount, turnover rate, utilization / productivity, compliance incident log, credentialing status, referral source breakdown, AR aging by payor
B2B SaaS	NRR, GRR, logo churn, CAC payback period, Rule of 40, ARR per FTE, magic number
Industrial / Mfg	Backlog (months), capacity utilization, on-time delivery %, aftermarket revenue %, inventory turns, CapEx / revenue
Consumer / DTC	12-month repeat rate, contribution margin, return rate, LTV/CAC, blended CAC trend, channel mix, platform concentration

6.4 Output contract

- KPI table: metric name, stated value, flag vs. threshold (Green/Yellow/Red), confidence, source.
- Missing KPI list: expected metrics not found in VDR — formatted as management question items.
- Delivery model flag (tech services): contractor %, geography concentration.

7. Legal & Contracts Agent (Phase 3 · Guideline 5)

ℹ️ Primary sources: all documents tagged LEGAL. Also receives contract review triggers from the Customer Quality Agent for material-customer contracts. Does not provide legal advice — produces a structured contract register for outside counsel review.

7.1 Triggered reviews vs. routine reviews

The Legal Agent performs two types of review:

- **Triggered:** Any contract identified by the Customer Quality Agent as belonging to a customer >20% of revenue. Reviewed immediately (Priority Tier equivalent within Phase 3). Austin: 'If there is a contract, is it [possible to] terminate it upon a change of control? Like anything that's going to affect the deal.'
- **Routine:** All other contracts tagged LEGAL. Processed after triggered reviews.

7.2 Term extraction schema (per contract)

Field	What to extract	Risk flag rule
Change-of-control clause	Required / not required; consent standard; trigger threshold (% ownership change)	Affirmative consent required + customer >20% revenue = Red
Termination for convenience	Yes/No; notice period; any penalty	Notice <60 days, no penalty + customer >20% = Red
Anti-assignment	Yes/No; whether it captures CoC	Captures CoC + material counterparty = Yellow
Exclusivity / MFN / non-compete	Scope, geography, duration	Any exclusivity or non-compete limiting add-on M&A = Yellow
Auto-renewal / renewal mechanics	Yes/No; notice period for non-renewal	No auto-renewal + >10% customer = Yellow
Pricing terms / escalators	Fixed, indexed, customer-side reset rights	Customer-side repricing without floor = Yellow
Liability cap / indemnity	Cap amount; carve-outs	Uncapped liability or unusual indemnity scope = Yellow/Red
Data / IP / privacy	Key obligations; IP ownership transfers	Unusual obligations — pass to outside counsel

7.3 Cross-contract roll-ups (output for Cross-Analysis Agent)

- CoC consent list — every contract requiring consent, ranked by revenue impact.
- Termination exposure — total revenue under short-notice termination-for-convenience contracts.

- Restrictive covenant map — every exclusivity, non-compete, MFN the company has agreed to.
- Litigation register — all open matters, demand letters, regulatory correspondence.

7.4 Output contract

- Contract register: one row per contract, all fields per §7.2, risk flag per clause.
- CoC consent list (high-priority deliverable to outside counsel).
- Termination exposure total ($).
- Litigation and compliance register.
- Data room gaps: material customers with no contract found.

8. Quality of Earnings Agent (Phase 3 · Guideline 6)

ℹ️ Primary sources: QofE report (if present), EBITDA bridge, addback schedule passed from the Financial Trends Agent. Purpose: structured flag log to feed the formal QofE workstream. Not a replacement for the accounting provider.

8.1 Addback tiering (primary framework)

Tier	Description	Examples
Tier 1 — Strong	Clearly one-time, documented, no recurring substitute	Closed legal settlement, facility relocation, one-time IT migration
Tier 2 — Defensible	Owner/private-company costs that won't recur under PE ownership — need corroboration	Above-market owner comp (vs. comp survey), owner personal expenses, family members on payroll with no operational role
Tier 3 — Stretch	Defensible but require execution risk post-close	Run-rate adjustments for hires not yet made, pro-forma synergies, cost cuts where role may need replacing
Tier 4 — Reach	Unlikely to survive buyer QofE; flag for immediate management question	Recurring marketing spend labelled one-time, growth investments labelled addbacks, COVID/stimulus without contemporaneous docs

ℹ️ Rule: any addback without supporting documentation referenced in the schedule is auto-classified Tier 4. Any single addback >5% of reported EBITDA should have documentation in the VDR — flag if absent.

8.2 Revenue quality flags

Flag	Detection rule
Revenue tied to episodic / event-driven demand (Austin, healthcare)	Flag if volume is hard to forecast, referral patterns inconsistent, or revenue depends on discrete events
Revenue recognized before delivery or cash collection	DSO trending up >10 days over trailing 12 months; any bill-and-hold language in revenue notes
Period-end revenue spike	Last week of period revenue >2x weekly average if monthly data available
Non-recurring items in run-rate	Any large one-off contract included in TTM run-rate without normalization
Addbacks growing faster than EBITDA	Mgmt-adjusted EBITDA growing faster than reported EBITDA over 3 years — flag for QofE provider
Policy change in revenue recognition	Footnote difference between current and prior-year audited statements

8.3 Output contract

- Addback ledger: one row per addback, tier classification, supporting doc reference (or 'Not found'), estimated dollar impact.
- Revenue quality flag log: each flag with evidence and source citation.
- Pre-QofE scope items: structured list of issues for the accounting provider to validate.
- Adjusted EBITDA range: reported → Tier 1+2 only → Tier 1 only (three versions the deal team can use for scenario modeling).

9. Forecast Agent (Phase 3 · Guideline 7)

> ℹ️ *Primary source: financial model / projection file (Priority Tier). If no model: CIM forward projections + management presentation. Phase 1: treat stated assumptions as source of truth; note where assumptions are unsupported by data present in the VDR.*

9.1 Forecast credibility rubric

Rating	Definition	Examples
Supported	Consistent with trailing 3-year performance OR backed by contracted revenue / signed pipeline / documented price change	Revenue growth assumes 90% from existing customer base on contracted ARR
Plausible	Achievable but a step-change from trailing performance — named driver exists	New product line contributing 20% in Y2 — mentioned in CIM but not yet in revenue
Stretch	Requires sustained performance materially outside historical envelope, OR contradicted by recent trends	EBITDA margin expanding 800bps over 3 years when historical trend is flat

9.2 Forecast assumptions to extract and evaluate

Assumption	What to check
Revenue growth rate	Compare implied YoY growth to trailing 3-year actuals. Is the forecast above, below, or in line?
Gross margin improvement	Implied margin expansion vs. historical trend. Is it supported by a named driver (pricing, mix)?
New customer revenue	What % of forecast revenue is from new customers? Does the pipeline file support this?
Hiring plan	Implied headcount growth vs. revenue growth. Does the implied revenue per FTE hold up vs. history?
Backlog / pipeline coverage	Does stated pipeline × historical conversion rate ≥ forecast new-customer revenue? (Austin: backlog and pipeline are key for tech services)
CapEx and WC assumptions	Any improvement assumptions for DSO, DPO, or capex ratio? Flag if assumed vs. demonstrated.

9.3 Downside sensitivity inputs

The Forecast Agent generates the following sensitivity inputs for the deal team's model:

- Top-customer loss case: impact of losing each of the top 3 customers individually (revenue and EBITDA impact).
- Growth rate haircut case: what if growth tracks trailing 3-year average instead of forecast?
- Margin compression case: trailing 3-year average gross margin instead of projected expanded margin.
- Addback erosion case: EBITDA with only Tier 1 + Tier 2 addbacks (from QofE Agent output).
- Pipeline miss case: new-customer revenue at 50% of forecast (if pipeline coverage is <2x).

9.4 Output contract

- Forecast assumption credibility table: each assumption rated Supported / Plausible / Stretch.
- Revenue build comparison: forecast vs. trailing actuals, delta, flag.
- Downside sensitivity inputs (for deal team LBO model).
- Management validation items: assumptions requiring management to substantiate.

10. Cross-Analysis Agent (Phase 4 · Guideline 8)

ℹ️ Runs after all seven Phase 3 workstream agents complete. Purpose: (1) reconcile findings across workstreams; (2) check CIM claims against data room evidence; (3) rank and prioritize flags; (4) produce the top 10 issues list.

10.1 Cross-workstream reconciliation checks

These checks are the most distinctive capability of the multi-agent system — each requires output from at least two different workstream agents. Phase 1: the agent notes mismatches but presents them as questions, not verdicts.

Check	Agents involved	How it works
CIM revenue claim vs. audited/stated financials	Business Model + Financial Trends	CIM says revenue is $X. Financial Trends Agent extracted $Y from financials. Flag if delta >1%.
CIM customer concentration claim vs. customer workbook	Business Model + Customer Quality	CIM says 'no customer >15%'. Customer workbook shows 28%. Flag mismatch.
Stated NRR vs. GRR — sanity check	Customer Quality + QofE	NRR should always be ≥ GRR. If NRR < GRR as stated, flag as metric error.
Customer concentration trigger → contract review	Customer Quality → Legal	Customer >20% identified; Legal Agent confirms whether contract was found and reviewed.
Revenue model claim vs. contract terms	Business Model + Legal	CIM claims 'all customers on multi-year contracts.' Legal Agent found contracts with annual auto-renew only. Flag mismatch.
EBITDA addback vs. supporting docs	Financial Trends + QofE	Addback schedule lists $X for owner comp. QofE Agent notes no comp survey in VDR. Flag.
Forecast new-customer revenue vs. pipeline coverage	Forecast + KPI	Forecast assumes $Xm from new customers. KPI Agent extracted pipeline of $Ym at Z% conversion. Does $Y × Z ≥ $X?
Stated headcount vs. implied from financials	Financial Trends + KPI	Payroll cost ÷ stated avg comp = implied headcount. Does it match stated headcount?
Stated DSO vs. AR ÷ (revenue/365)	Financial Trends	Internal consistency check within Financial Trends Agent output.
Healthcare compliance claims vs. compliance file	Business Model + Legal + KPI	CIM states 'no material compliance issues.' Legal Agent found open regulatory correspondence. Flag.

10.2 CIM claims vs. data room

The Cross-Analysis Agent maintains a structured list of every factual claim in the CIM and checks it against the data room evidence. Any claim that cannot be corroborated is flagged. This list is one of the most useful deliverables for the management meeting.

10.3 Flag prioritization and top 10 issues

Severity	Definition
Critical	Could change LOI price or structure, or block close. Examples: material customer with CoC veto, EBITDA addback >10% Tier 4, undisclosed litigation, compliance incident.
Material	Needs to be understood; may shape underwriting. Manageable with information. Examples: concentration at 22%, margin below threshold but improving.
Track	Worth noting and asking about; not deal-shaping. Examples: below-benchmark average account size; no formal KPI dashboard.

10.4 Output contract

- Reconciliation log: every cross-check run, result (match / mismatch / cannot check), materiality.
- CIM claims vs. data room list: claim, evidence found, status.
- Top 10 diligence issues: ordered by severity × deal relevance, each with source citations.
- Data room gap list: all expected documents not found, formatted as information request.

11. Orchestrator Agent (Phase 5 · Guideline 9)

ℹ️ *The Orchestrator is the only agent whose primary job is output generation. It takes the structured outputs from all prior agents and assembles the final deliverables. It validates coherence, assigns final section-level confidence, and formats outputs per the templates below.*

11.1 Coherence validation before output

Before generating output, the Orchestrator runs a coherence check — ensuring that outputs from different agents do not contradict each other without explanation:

- A Red flag in Financial Trends must be reflected in the executive summary.
- If the Legal Agent found a CoC issue, it must appear in the top 10 issues from the Cross-Analysis Agent.
- Confidence ratings across sections must be internally consistent (if source documents are few, no section should claim High confidence).
- Source citations must resolve — every citation references a document that exists in the indexed store.

11.2 Output deliverables

Deliverable	Format	Primary use case
Diligence memo	Word (.docx)	IC pre-read; replaces the manual associate memo. Structure: Exec Summary → Section ratings grid → Sections 1–7 → Appendices.
Executive deck	PowerPoint (.pptx)	IC meeting. ~15 slides: deal at a glance, diligence ratings grid, top 10 issues, top 10 questions, financial summary, concentration & retention, contract summary, forecast assessment, risks & mitigants, value creation levers.
One-page company overview	PDF	Senior partner screen; pre-LOI. Single page: business description, key financials, customer snapshot, KPI snapshot, top 5 risks, top 5 levers.
Risks & mitigants grid	Word table or PPT slide	Risk × severity × evidence × mitigant × owner × status.
KPI dashboard	PDF or PPT	Visual: KPI name, company value, benchmark bar, trend sparkline, flag.
Management question list	Word (.docx)	Top 10 questions each: for management, for banker, for advisor.
Reconciliation log	PDF	Full log for QofE provider and outside counsel kickoff.
Data room gap list	PDF	Formatted as an information request to the seller/banker.
Diligence tracker	Working doc (Word or Excel)	Living document tracking every open flag with owner, status, resolution.

11.3 Section-level confidence assignment

Confidence	Trigger
High	Primary source (audited financials, signed contract, structured database extract); corroborated by ≥1 other document; extraction confidence high.
Medium	Single source or CIM only; corroboration partial; or primary source with medium extraction confidence.
Low	Narrative or unsourced claim; contradicted by another document; OR data missing and conclusion is inferred. Non-banked deal automatically reduces all sections to Medium until financials confirmed.

11.4 Source citation format

Every claim in every deliverable: "[Claim]. ([Document name], [page/tab/cell], [extraction confidence])"

Example: "Revenue grew 22% in FY2024, driven primarily by existing-customer expansion. (CIM, p.14 — High confidence; corroborated by Financial Trends Agent extraction from Management Accounts Dec 2024, tab 'Monthly P&L' — High confidence.)"

12. Implementation Roadmap

Sequencing reflects the agreed sprint plan from the Rallyday / Nimble Gravity working session. Phase 1 and Phase 2 (ingestion and classification) are substantially complete. Work now begins on the Phase 3 workstream agents, starting with the Financial Trends Agent.

12.1 Current state (as of May 2026 working session)

- Phase 1 — Ingestion Agent: complete. Can ingest any SharePoint or folder source, traverse structure, and store indexed documents. Processing 1,200 documents takes approximately 2 hours.
- Phase 2 — Classification: in progress. Document classification engine is being built. Company profiler is next.
- Phase 3 onward: not yet started. Financial Trends Agent is the immediate next build.

12.2 Sprint 2 target

- Financial Trends Agent: extract stated revenue, gross margin, and EBITDA from Priority Tier documents. Apply Austin's primary thresholds for tech services and healthcare. Produce structured output.
- Initial output for Rallyday review: show that the system can ingest, classify, and extract financial metrics from the elder care data room, and surface flags against the primary thresholds.
- UI: work with Matt Crysler in parallel to design the output presentation layer.

12.3 Build sequence for remaining agents

Sprint	Agent(s)	Rationale
2	Financial Trends Agent	Highest priority per Austin; delivers first visible output to Rallyday
3	Customer Quality Agent + Business Model Agent	Concentration and model quality are the next most requested outputs; leverage Priority Tier document extraction already done
4	KPI Agent + Quality of Earnings Agent	KPI requires overlay confirmation; QofE needs Financial Trends output
5	Legal & Contracts Agent + Forecast Agent	Legal requires triggered inputs from Customer Quality; Forecast needs Financial Trends and KPI
6	Cross-Analysis Agent	Requires all Phase 3 agents to be stable
7	Orchestrator Agent + full output generation	Final assembly; requires all prior agents stable

12.4 Document expansion — additional data rooms

Austin confirmed he will share 2–3 additional company data rooms (starting with Simplistic). The build team will use these to:

- Map common language and document naming conventions across data rooms — so the Ingestion Agent's Priority Tier detection rules work reliably across different bankers and deal teams.
- Validate that the extraction schema handles different financial statement formats (e.g., P&L as PDF vs. as Excel model).
- Stress-test the classification engine on diverse document sets.

12.5 Deferred to Phase 2+ (not in current scope)

- Cross-document reconciliation (recomputing NRR from raw cohort data, recomputing DSO from AR/revenue): Phase 2. Austin confirmed Phase 1 should treat stated figures as source of truth.
- Composite deal scoring: explicitly excluded, per v1 decision maintained in v2.
- Voice-of-customer synthesis from call transcripts.
- Market sizing and competitive landscape.
- 100-Day plan generation.

Appendix A — Primary Threshold Reference (Austin Hough, Rallyday Partners)

The thresholds below are the primary evaluation criteria for this system. Source: Austin Hough email to Boris Iantchoulev et al., May 20, 2026 (subject: 'RE: NG | Rallyday - UC13 Weekly Checkpoint'). Reproduced here as the authoritative reference for all agents.

Tech Services Businesses — Primary Thresholds

Metric	Flag threshold	Austin's note
Net Revenue Retention	Red: <~90%	Suggests churn pressure, weaker customer stickiness, or a more project-based / transactional model
Gross Revenue Retention	Red: <~80–85%	Explicit threshold
Gross Margin	Red: <~40% (premium digital services: 45–50%+ preferred)	Could indicate lower-value work, weak utilization, poor delivery leverage, or insufficient pricing power
Top customer concentration	Red: >~25% of revenue	Flag as concentration risk, particularly if project-driven, contractually uncommitted
Organic revenue growth	Red: <~10–15%	For a premium tech services platform — raises questions on positioning, sales engine, customer demand
EBITDA margin	Yellow: <~10–15%	Depends on stage — may be fine, but should be flagged
Average account size	Yellow: <~$100–200K	Smaller accounts imply SMB exposure, lower budget durability, harder path to scale
Revenue model	Yellow: mostly one-time project work	Not necessarily a deal-killer, but usually deserves a flag
Delivery model	Yellow: contractor-heavy or single-geography (especially India)	Delivery model risk

Healthcare Services Businesses — Primary Thresholds

Metric	Flag threshold	Austin's note
Revenue growth	Red: <~5–10%	Low growth may be fine in stable markets but should prompt questions on referral trends, reimbursement, competition, provider / caregiver availability
EBITDA margin	Red: <~10%	Varies by subsector — sub-10% may indicate labor inefficiency, weak reimbursement, poor scheduling / utilization, high admin burden, or lack of scale
Gross margin	Red: <~30–35%	For people-heavy services: indicator of wage pressure,

Metric	Flag threshold	Austin's note
		poor labor utilization, pricing constraints, or unfavorable payor / customer mix
Customer / referral concentration	Red: >~20–25% from single referral source, facility, health system, payor, or customer	Treat identically to customer concentration in tech
Government / payor concentration	Yellow: >~50% Medicare, Medicaid, VA, or small number of managed care / government-related payors	Flag if heavily dependent on government reimbursement
Episodic / event-driven revenue	Yellow: flag if volume hard to forecast, referral patterns inconsistent, or depends on discrete events	Similar to project revenue in tech
Employee turnover	Red: >~30–40%	High turnover creates quality issues, hiring costs, capacity constraints, compliance risk, and margin pressure
Recruiting funnel	Yellow: flag if constrained	Constrained recruiting can cap growth even if current revenue looks good
Utilization / productivity	Yellow: flag low utilization for clinic-based, home care, or field services models	Major margin driver
Compliance / quality issues	Red: any history of audits, adverse survey findings, licensing issues, litigation, credentialing gaps, billing / coding concerns	Any history — not just open matters
Site-level visibility (multi-site)	Yellow: flag if cannot produce revenue, margin, volume, utilization, quality at location level	Inability to produce this is itself a management capability issue

Appendix B — Secondary Benchmark Reference (Public Sources)

These benchmarks from publicly available sources (SaaS Capital, Benchmarkit, Damodaran / NYU Stern, Service Performance Insight, GF Data / Capstone Partners) are retained as supplementary reference. They are used when: (a) a deal falls outside the two primary overlay industries; or (b) Austin's primary thresholds do not cover a specific metric needed for a non-tech, non-healthcare deal.

Refer to the v1 specification (Section 2 through Section 12) for the full public-source benchmark library. These are not reproduced here to avoid confusion with Austin's primary thresholds, but remain available in v1 and in the agent configuration layer.

Appendix C — Glossary

Term	Definition
CIM	Confidential Information Memorandum. The primary marketing document prepared by the seller's investment bank. The most information-dense single document in a banked process.
QofE / Q of E	Quality of Earnings report. A third-party accounting firm report that validates historical EBITDA and identifies addbacks. Historic-facing only (no projections).
NRR	Net Revenue Retention. Existing customer ARR at end of period (including expansion, less churn and contraction) ÷ existing customer ARR at start of period. Always ≥ GRR.
GRR	Gross Revenue Retention. Existing customer ARR retained (excluding expansion) ÷ starting ARR. Isolates pure churn.
CoC	Change of Control. A clause in a contract defining what happens when ownership of a party changes. Can require consent, allow termination, or accelerate obligations.
LBO	Leveraged Buyout. A control acquisition financed with a significant proportion of debt.
TTM / LTM	Trailing Twelve Months / Last Twelve Months. Same concept — the most recent 12-month period of financial performance.
Priority Tier	The ~5–10 VDR documents that contain ~90% of diligence-relevant signal, per Austin's guidance. Processed first by the Ingestion Agent.
Workstream agent	A Phase 3 agent responsible for a specific analysis domain (Financial Trends, Customer Quality, etc.). Each produces a structured output per its output contract.
Cross-Analysis Agent	The Phase 4 agent that reconciles outputs across all workstream agents, checks CIM vs. data room, and produces the final ranked flag list.
Orchestrator Agent	The Phase 5 agent that assembles the final deliverables from all prior agent outputs.